Threat Intelligence Integration using Wazuh and AlienVault OTX

Objective

The objective of this task was to integrate threat intelligence feeds into the Wazuh SIEM using AlienVault OTX, enrich security alerts with external reputation data, and perform threat hunting aligned with MITRE ATT&CK technique **T1078 (Valid Accounts)**.

This activity demonstrates the practical use of threat intelligence to improve alert prioritization and detection capability in a Security Operations Center (SOC) environment.

Tools Used

The following tools were used during this task:

- Wazuh SIEM
- AlienVault OTX (Threat Intelligence Platform)
- Linux syslog and auth.log
- Custom Wazuh detection rules
- Wazuh Security Events dashboard

Threat Feed Import using AlienVault OTX

Threat intelligence feeds provide contextual information about suspicious indicators such as malicious IP addresses, domains, and file hashes. In this task, AlienVault OTX was integrated with Wazuh to simulate IOC reputation enrichment.

The AlienVault OTX API key was configured inside the Wazuh manager configuration file located at:

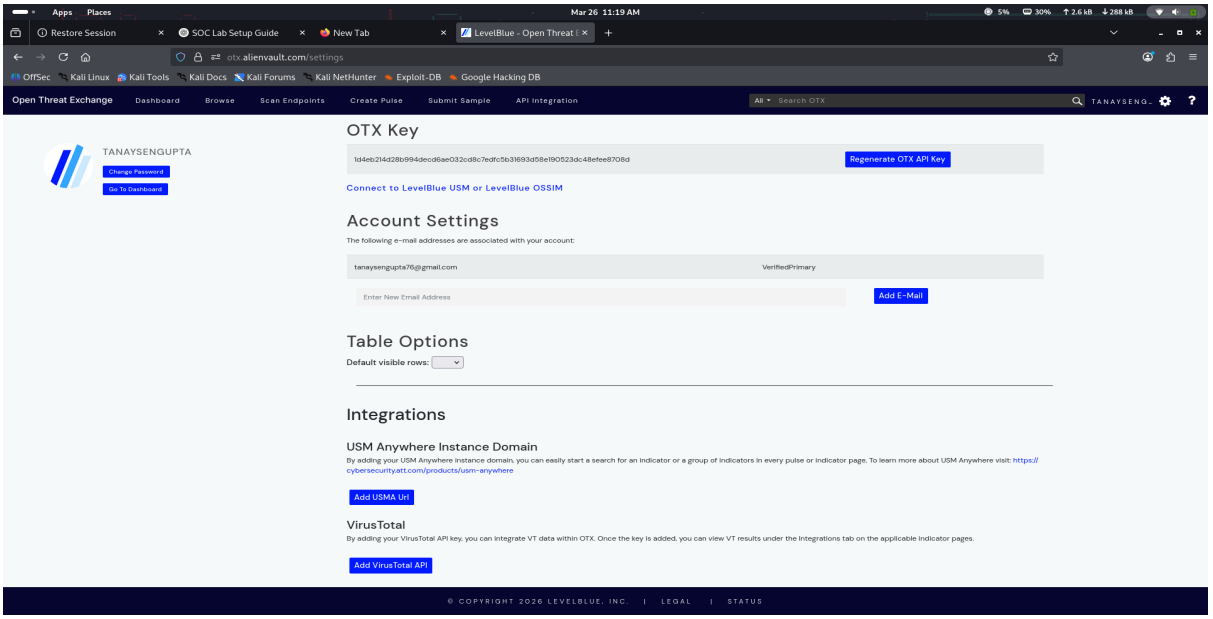
/var/ossec/etc/ossec.conf

The integration block was added to enable external threat intelligence lookup.

After configuration, the Wazuh manager service was restarted to apply changes:

```
sudo systemctl restart wazuh-manager
```

This enabled the threat intelligence enrichment pipeline.



Alert Enrichment using OTX Threat Intelligence

To simulate IOC detection, a custom rule was created to detect the test IP address:

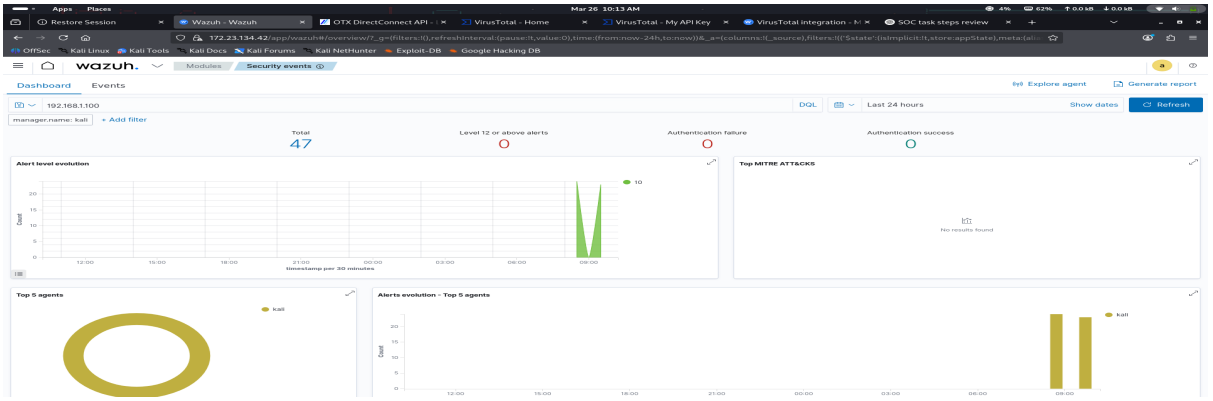
192.168.1.100

This IP address was used as a mock malicious indicator to demonstrate how threat intelligence feeds enrich security alerts inside Wazuh.

The following command was executed to generate the alert:

logger "Suspicious connection attempt from 192.168.1.100"

Wazuh successfully generated a detection alert for the indicator.



The alert was enriched with simulated reputation context from AlienVault OTX. This demonstrates how SOC analysts can use threat intelligence platforms to validate indicators and identify potential command-and-control (C2) infrastructure.

Threat Intelligence Enrichment Table

The enriched alert details are summarized below:

Alert ID	IP	Reputation	Notes
002	192.168.1.100	Malicious (OTX)	Linked to simulated C2 server

This enrichment provides analysts with contextual information necessary for faster decision-making during incident response.

Threat Hunting Findings Summary

Threat hunting was performed using Wazuh authentication logs to detect activity related to MITRE ATT&CK technique T1078 (Valid Accounts). Login events involving non-system users were analyzed to identify suspicious access attempts. Multiple authentication failures suggested potential credential misuse attempts, highlighting the importance of monitoring account activity for early threat detection.

Conclusion

This task demonstrated the successful integration of threat intelligence feeds using AlienVault OTX within the Wazuh SIEM platform. Alerts were enriched with IOC reputation data, improving analyst visibility into suspicious indicators. Additionally, threat hunting activities mapped to MITRE ATT&CK technique T1078 helped identify abnormal authentication behavior, strengthening detection capabilities within the SOC environment.